Flood Monitoring and Risk System: Hazard Definition, Multi-Source Validation, and Ingestion Data Contract

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Abstract

This document formalizes Workshop No. 1 of the semester-long course project in Data Analysis Programming. The objective is to design, validate, and contract the architectural foundation for an end-to-end multi-source data ingestion and risk analysis system dedicated to flood monitoring across the state of Texas, United States. We integrate four official data sources spanning historical archives (NOAA Storm Events Database and NOAA GHCN-Daily precipitation) and near-real-time streaming feeds (National Weather Service ASOS station observations and active weather alerts). We document the physical phenomenon of flash and riverine flooding, evaluate empirical update behaviors, perform comprehensive exploratory data profiling across more than 3,500 real records, and expose five concrete data-quality anomalies with authentic record fixtures. Furthermore, we establish a robust canonical event model with dedicated hydrological domain tables, prove cross-source event overlap across five real-world Texas flood occurrences, formalize an idempotent ingestion contract, and propose a calibrated Flood Risk Index (FRI).

Keywords: Flood monitoring, data engineering, canonical schema, data contract, Texas hydrometeorology, NOAA, National Weather Service, data quality profiling.

1. Project Overview and Hazard Definition

1.1. Physical Phenomenon and Regional Vulnerability

Flooding constitutes one of the most recurrent and economically catastrophic natural disasters in North America. The hydrometeorological profile of the State of Texas (Federal FIPS code 48) presents an extraordinary exposure profile driven by the atmospheric convergence of moisture-laden tropical maritime air masses from the Gulf of Mexico against the pronounced topographic uplift of the Balcones Escarpment (Slater et al., 2021). This geographic region, colloquially designated as *"Flash Flood Alley"*, exhibits some of the highest rainfall rate intensities globally, producing severe runoff events within minutes to hours (Bedient et al., 2019).

Floods within Texas manifest across three primary hydrological regimes:

1. **Flash Flooding:** Rapid surface runoff occurring when torrential convective rainfall rates overwhelm soil absorption capacities and urban storm-sewer networks, resulting in life-threatening stream crests and submerged low-water crossings.

2. **Riverine / Fluvial Flooding:** Slower-onset, prolonged inundations occurring when regional drainage basins (e.g., Brazos, Trinity, Colorado, and San Jacinto river systems) receive excessive precipitation over multi-day periods, causing rivers to burst their banks.

3. **Coastal / Surge Flooding:** Tropical cyclone storm surges driving ocean water inland along the low-lying Gulf coastal plain, frequently combined with severe freshwater drainage backwater effects in metropolitan Houston and Beaumont.

1.2. Severity Metrics and Measurement Units

Hazard severity is monitored through multiple quantitative and qualitative indicators:

- **Precipitation Accumulation:** Physical liquid precipitation measured in millimeters (*mm*) or inches (*in*) accumulated over discrete temporal durations (1-hour, 3-hour, 24-hour, and 7-day windows).

- **Hydrometric River Crest:** Stream stage height measured in feet or meters relative to established national flood action stages (Action, Minor, Moderate, Major Flood stages).

- **Socio-Economic Impacts:** Estimated monetary property and crop damages in United States Dollars (USD), direct human fatalities, injuries, and emergency evacuation counts.

- **Alerting Classifications:** Categorical National Weather Service (NWS) hazard tiers: *Minor, Moderate, Severe, and Extreme*.

1.3. Geographic and Temporal Scope

The spatial scope is bounded by the State of Texas, with focused analytical monitoring across four metropolitan epicenters and adjacent drainage corridors: Harris County (Houston metropolitan area and coastal bayous), Travis County (Austin and Colorado River basin), Bexar County (San Antonio and San Antonio River basin), and Dallas/Tarrant counties (Trinity River basin).

The historical analysis baseline spans six full calendar years from **2019 through 2024**, ensuring a statistically representative multi-year distribution satisfying the course requirements. The operational near-real-time situation is characterized by a rolling 24-hour window for automated surface station observations (ASOS), continuous active alert feeds, and a 7-day antecedent precipitation index.

1.4. Scope Exclusions and Technical Constraints

This data engineering system explicitly excludes continuous 2D hydrodynamic shallow-water differential equation simulations (e.g., HEC-RAS or hydraulic numerical mesh modeling), microscopic subsurface municipal pipe network modeling, individual parcel-level insurance claim underwriting, and multi-decadal climatological projections. The technical pipeline operates strictly upon open, non-authenticated federal data services, adhering to NOAA/NWS API usage guidelines.

2. Functional User Profile and User Stories

2.1. Target Functional User

The primary beneficiary of this platform is the **Emergency Management Analyst / Hydrological Hazard Coordinator** operating within civil protection agencies such as the Texas Division of Emergency Management (TDEM), the Harris County Flood Control District (HCFCD), or regional municipal Emergency Operations Centers (EOC).

This user is tasked with continuous threat assessment, real-time situational awareness, prepositioning high-water rescue assets, coordinating roadway closures at inundated low-water crossings, and briefing municipal directors regarding evacuation thresholds.

2.2. User Stories Prioritized via MoSCoW

Table 1 specifies six prioritized user stories structuring system capabilities across real-time monitoring, historical analysis, and operational data trust.

Table 1
System User Stories Prioritized via the MoSCoW Methodology

ID	Priority	User Story	Acceptance Criteria
US-01	Must Have (Monitoring)	*As an emergency analyst, I want to* view active NWS Flood and Flash Flood warnings on an interactive map *so that* I can identify counties facing imminent inundation.	Visualized on dashboard within 5 minutes of issuance, color-coded by severity with expiration timestamps.
US-02	Must Have (Monitoring)	*As an operations coordinator, I want to* inspect live precipitation rates (1h, 3h, 24h) from airport ASOS stations *so that* I can detect rapid flash-flood conditions.	Real-time observation table updated sub-hourly; highlights rain rates exceeding 25 mm/h.
US-03	Must Have (Analysis)	*As a hazard researcher, I want to* benchmark daily rainfall against 5-year GHCN-Daily historical distributions *so that* I can identify statistically anomalous deluges ($>$ 95th percentile).	Historical percentiles computed per station and month; anomaly badge displayed when exceeded.
US-04	Should Have (Analysis)	*As a civil protection director, I want to* consult an automated Flood Risk Index (LOW / MED / HIGH / CRITICAL) *so that* response resources are prioritized objectively.	Heuristic index combining rainfall, alert tier, and recurrence score, recalculated every cycle.
US-05	Should Have (Data Trust)	*As a data engineer, I want to* monitor watermark timestamps, run metrics, and quarantined rejected records *so that* I can guarantee data freshness and audit pipeline integrity.	Operational health panel displaying last poll timestamp, row counts, and raw error payloads.
US-06	Could Have (Analysis)	*As a risk analyst, I want to* correlate historical property damage figures with recorded rainfall totals *so that* I can estimate monetary vulnerability thresholds.	Historical query interface linking NOAA Storm Events damages with GHCN daily precipitation.

3. Four Validated Data Sources

All data sources were directly queried via HTTP protocols without third-party wrapper libraries. Raw payloads were preserved in the immutable landing zone `workshop_1/samples/`. Table 2 summarizes the technical parameters of the selected sources.

Table 2
Technical Characterization of Validated Hazard Data Sources

Attribute	NOAA Storm Events	GHCN-Daily (NCEI)	NWS Observations	NWS Active Alerts
Source Type	Historical Archive	Historical Baseline	Incremental Feed	Incremental Feed
Format	Bulk GZipped CSV	REST JSON	GeoJSON	GeoJSON / CAP
Endpoint / URL	NCEI SWDI Portal	NCEI Data Service v1	`/stations/{id}/obs`	`/alerts/active?area=TX`
Authentication	Public Domain	Public Domain	Descriptive User-Agent	Descriptive User-Agent
Sample Size	582 records (TX 2024)	2,188 observations	500 records	253 alert features
Update Policy	Monthly recompilation	Daily (24h latency)	Sub-hourly (METAR)	Continuous event-driven
Rate Limits	Local caching	Polite batching	Throttled bursts	5–10 min polling

3.1. Empirical Update Behavior Verification

Empirical characterization was conducted by executing repeated queries separated in time:

- **NWS Station Observations:** Two queries executed 60 minutes apart for station KHOU (Houston Hobby Airport) confirmed that the observation timestamp advanced from `2026-11-05T18:40:00` to `2026-11-05T19:40:00Z`, updating ambient temperature, pressure, and wind speed.
- **NWS Active Alerts:** Two queries executed 90 minutes apart demonstrated that short-fused urban advisories expired and were automatically purged from the active feed, while new convective statements were appended.
- **Watermark Implication:** Because weather alerts are mutable entities transitioning between states (*Actual, Update, Cancel*), the watermark must track `properties.sent` with a 2-hour sliding overlap window to capture late updates and explicit cancellations.

4. Source Profiling and Data Quality Findings

4.1. Profiling and Exploratory Findings

Execution of the complete profiling notebook (`notebooks/01_source_profiling.ipynb`) revealed high geographic integrity: 100% of latitude and longitude coordinates for Texas flood events in NOAA Storm Events fell strictly within Texas bounding coordinates ($[25.8°\text{N}, 36.5°\text{N}]$, $[-106.6°\text{W}, -93.5°\text{W}]$).

4.2. Five Concrete Data-Quality Anomalies

In accordance with course evaluation criteria, five distinct data quality problems were discovered in the authentic raw data:

1. **Whitespace Padding and String Typings in GHCN-Daily:** Numeric precipitation and temperature values are returned as fixed-width, whitespace-padded strings (e.g., `'  104'`, `'    0'`). Direct arithmetic in Python raises unhandled `TypeError` exceptions.

2. **Implicit Unit Scaling in GHCN-Daily:** Precipitation is reported in tenths of a millimeter without unit metadata in the JSON payload (e.g., station USW00012921 on 2024-01-22 recorded `PRCP: '  612'`). Ingesting this value without dividing by 10.0 inflates rainfall by 1,000% (61.2 mm versus 612 mm).

3. **Fragmented Non-UTC Timestamps in NOAA Storm Events:** Event start times are distributed across four separate columns (`BEGIN_YEARMONTH`, `BEGIN_DAY`, `BEGIN_TIME`, `CZ_TIMEZONE`) expressed in local civil daylight saving codes (`CST-6`, `CDT-5`), necessitating custom parsing and offset correction to UTC.

4. **High Sensor Sparseness in NWS Observations:** Over 85% of NWS station records exhibit `value: null` for short-term accumulation metrics (`precipitationLast3Hours`). The ingestion contract must explicitly differentiate between zero rainfall and sensor unequipped/offline states.

5. **Spatial Representation Disparity in NWS Alerts:** 36.4% of alerts possess `geometry: null`, conveying geographic boundaries exclusively via lists of administrative UGC and SAME county codes. Direct spatial joins (`ST_Intersects`) fail unless administrative boundary polygons are resolved from an authoritative lookup table.

4.3. Cross-Source Overlap Verification

Cross-source event matching was proven by correlating NOAA Storm Events flood reports against daily rainfall observations recorded by GHCN-Daily weather stations on identical dates in Texas during 2024 (Table 3).

Table 3
Cross-Source Overlap Verification Between NOAA Storm Events and GHCN-Daily (Texas, 2024)

Date	County	Storm ID	Hazard Type	Coordinates	GHCN Station	GHCN Rain
2024-01-22	Bexar	1159212	Flash Flood	(29.48, -98.49)	USW00012921	61.2 mm
2024-01-24	Harris	1160174	River Flood	(30.10, -95.76)	USW00012918	48.3 mm
2024-01-24	Bexar	1159392	Flash Flood	(29.41, -98.61)	USW00012921	40.6 mm
2024-05-02	Harris	1188444	Flash Flood	(29.99, -95.28)	USW00012918	34.8 mm
2024-05-02	Harris	1188445	Flash Flood	(30.05, -95.46)	USW00012918	34.8 mm

Analysis of Disparities Between Sources:

- *Temporal Granularity:* Storm Events records exact minute-level disaster onset (e.g., `10:15:00 CST`), whereas GHCN-Daily provides an aggregated 24-hour total.
- *Spatial Resolution:* Storm Events pinpoints specific low-water road crossings or inundated subdivisions, whereas GHCN-Daily provides fixed coordinates of the regional airport installation.
- *Phenomenological Dimension:* Storm Events captures societal disruption and emergency responses, whereas GHCN-Daily quantifies the physical hydrometeorological forcing mechanism.

5. Canonical Data Model and Ingestion Contract

5.1. Canonical Event Abstraction

Heterogeneous observations are normalized into the canonical `event` entity, defined by:

$$\text{Event} = \langle \text{event_id}, \text{source}, \text{event_type}, \text{timestamp}_{\text{UTC}}, \text{latitude}, \text{longitude}, \text{severity}, \text{raw_reference} \rangle \tag{1}$$

Specialized attributes are partitioned into domain tables linked via foreign keys:

- `flood_event_details`**:** Foreign key to `event_id`, capturing meteorological flood causes, county FIPS codes, monetary property losses in USD, direct deaths, injuries, and disaster narratives.

- `precipitation_reading_details`: Foreign key to `event_id`, storing station identifiers, calibrated rainfall in millimeters, accumulation window hours, and sensor quality control flags.

5.2. Ingestion Contract and Idempotency Guarantee

- **Historical Bulk Strategy:** NOAA Storm Events is ingested through annual archive partitions (2019–2024). GHCN-Daily is queried in monthly intervals per station. Duplicate resolution relies on `ON CONFLICT DO NOTHING` based on `EVENT_ID` and composite keys `(STATION, DATE)`.
- **Incremental Strategy:** Polling every 30 minutes for NWS station observations and every 5–10 minutes for NWS alerts. Watermarks track `properties.timestamp` and `properties.sent` with a 2-hour overlap lookback window.
- **Failure Recovery:** Ingestion cycles execute inside atomic database transactions. Malformed records are isolated into the `rejected_record` quarantine table. The `ingestion_state.watermark` advances strictly upon transaction commit, ensuring zero duplicate generation and zero data loss if a run crashes halfway.

6. Flood Risk Index (FRI) Indicator Proposal

6.1. Formulation and Output Scale

The **Flood Risk Index (FRI)** is a normalized dimensionless metric between 0 and 100 representing regional hazard threat:

$$FRI = 0.45 \cdot I_{\text{precip}} + 0.35 \cdot I_{\text{alert}} + 0.20 \cdot I_{\text{hist}} \tag{2}$$

Where:

- $I_{\text{precip}} = \min\left(100, \frac{R_{24h}}{P_{95}} \times 100\right)$ benchmarks current 24-hour rainfall (R_{24h}) against the 95th historical percentile of daily precipitation (P_{95}) for that station.
- I_{alert} encodes active emergency tiers: None = 0, Flood Advisory/Watch = 35, Flood Warning = 75, Flash Flood Warning/Emergency = 100.
- I_{hist} scores historical county flood recurrence over 2019–2024 from NOAA Storm Events.

The computed index maps to four operational alert tiers:

- $0 \leq FRI < 40 \implies$ **LOW**: Routine conditions, normal drainage flow.
- $40 \leq FRI < 70 \implies$ **MEDIUM**: Elevated moisture, localized low-water ponding.
- $70 \leq FRI < 90 \implies$ **HIGH**: Significant flooding, road closures, emergency readiness.
- $90 \leq FRI \leq 100 \implies$ **CRITICAL**: Life-threatening flash flooding, mandatory evacuations.

6.2. Validation Strategy and Operational Disclaimer

The algorithm will be validated by back-testing historical events, specifically the May 2024 Southeast Texas floods (Harris and San Jacinto counties) and the May 2021 flash floods, confirming that the index reliably triggers $FRI \geq 85$ without generating false positives during dry periods. The FRI is strictly an **operational situational monitoring aid** and does not constitute a legal flood guarantee or deterministic hydraulic model.

7. System Architecture and Risk Mitigations

7.1. Technical Risk Mitigation Matrix

- **Risk 1 (NWS API Rate Limits):** Mitigated through exponential backoff retries (3 attempts), local payload caching, and mandatory descriptive User-Agent headers.
- **Risk 2 (Alert Spatial Geometry Disparity):** Mitigated by maintaining a pre-loaded GIS boundary reference table mapping UGC/SAME codes to county centroids.
- **Risk 3 (NOAA Storm Events Release Latency):** Mitigated by restricting NOAA archives to historical baseline generation while driving real-time analysis with NWS feeds.
- **Risk 4 (Sensor Sparseness / Null Rates):** Mitigated via multi-station inverse-distance spatial averaging and explicit null-coalescing logic.
- **Risk 5 (Memory Pressures during Bulk Ingestion):** Mitigated by chunked streaming decoders and batched database inserts ($1,000$ rows per commit).

7.2. Architecture Overview

The system deploys locally via Docker Compose, comprising a PostgreSQL/PostGIS relational database, a scheduled Python ingestion engine (`ingest.py`), an immutable landing zone (`data/raw/`), a data quarantine table (`rejected_record`), and an interactive Plotly Dash analytical dashboard operating at `localhost:8050`.

8. Conclusion

Workshop No. 1 establishes a verified, empirically tested foundation for the Team 2 Flood Monitoring System. All four data sources have been proven functional with authentic records, data quality anomalies have been documented with concrete fixtures, the canonical data model has been mapped to domain tables, and the technical contracts guarantee idempotency and fault tolerance for subsequent implementation workshops.

References

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